

(a) [10 marks] In **Figure 2**, when input is LOW ($V_i = 0 \text{ V}$), the total power dissipation in circuit is **1.2 mW**.

Calculate the current flowing through this logic circuit and **estimate** the value of R_i in $k\Omega$.

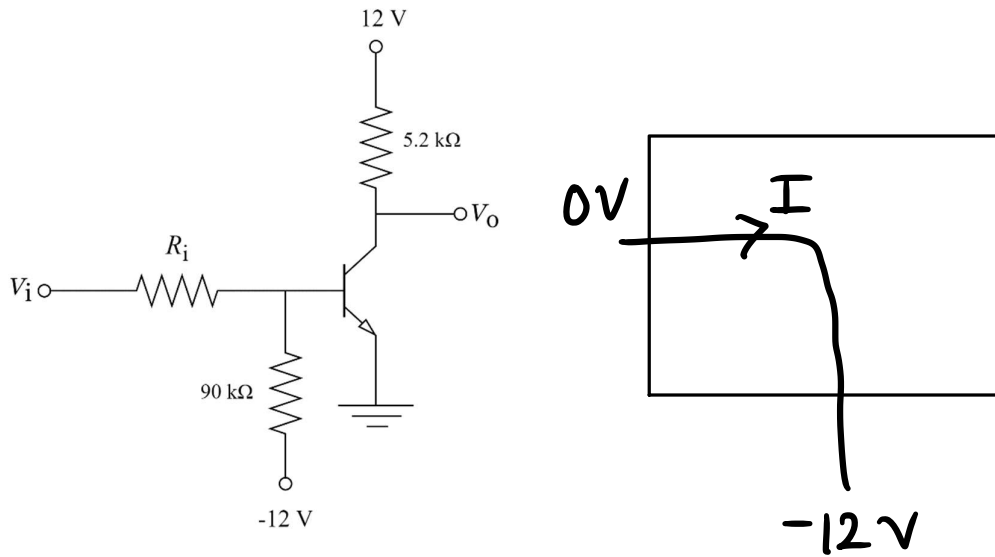


Figure 2

$$\begin{aligned}
 P &= \Delta V I \\
 1.2 \text{ mW} &= (0 + 12) \times I \\
 \Rightarrow I &= 0.1 \text{ mA}
 \end{aligned}
 \left| \begin{aligned}
 I &= \frac{0 + 12}{90 + R_i} \\
 \Rightarrow R_i &= \frac{12}{0.1} - 90 \\
 &= 30 \text{ k}\Omega
 \end{aligned} \right.$$